Empowering Compute, Network, Storage & AI

100 GbE / RoCE v2 NIC Benchmark & Validation Report

Broadcom NetXtreme-E BCM57508 & BCM57504 - back-to-back + through-switch

Server / Host	srv218 <-> srv148 (10.10.10.0/24 (back-to-back))
Platform	srv218 Tyrone Camarero (2x Xeon Gold 6338) <-> srv148 Tyrone MDA200A2N-224 (2x EPYC 9135)
Validation Tier	Qualification
Campaign Window	back-to-back + through-switch (09 Jun 2026)
Prepared For	Netweb Technologies India Limited
Date	2026-06-09

Overall Result: PASS

Executive Summary

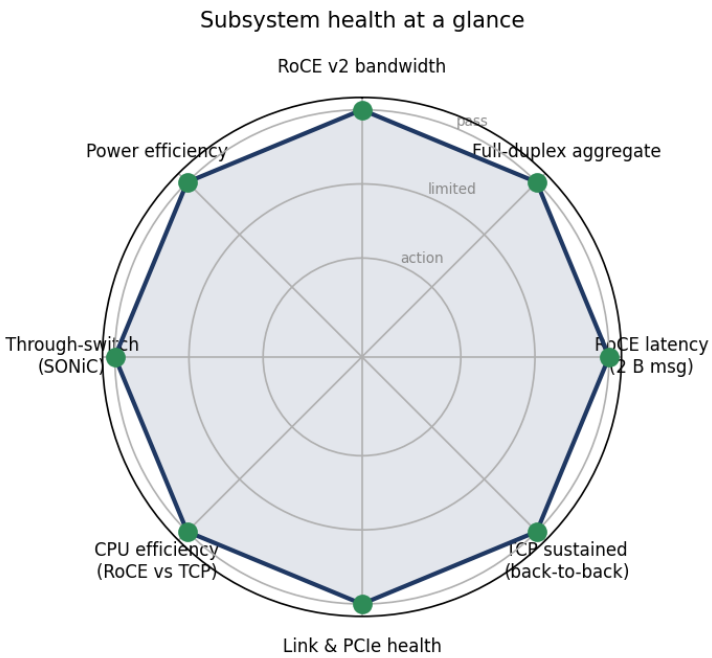
Validation of two Broadcom NetXtreme-E 100 G Ethernet adapters (BCM57508 / BCM57504) in Netweb / Tyrone Systems servers, connected back-to-back over a single 100 G DAC cable, then re-validated through a production-class Accton AS4630-54TE / SONiC switch. All measured numbers match or exceed the published Broadcom Thor envelope, with full line rate on RoCE v2 bandwidth and latency in the expected microsecond band.

The two NICs operate at full 100 G line rate for RoCE v2 send / write / read (98.18 Gb/s), sustain 194.99 Gb/s full-duplex, with no FCS / PCS / pause errors. RoCE uses ~9.2x less host CPU than TCP on the receive side (kernel bypass: %softirq drops to 0%).

Through the switch (default lossy, no PFC), RoCE held full line rate while TCP collapsed 94 -> 68 Gb/s from buffer overruns - and RoCE drew 27-49 W less per server than TCP at sustained 100 G. RoCE is robust through any reasonable fabric; TCP needs PFC/ECN tuning to stay healthy at 100 G.

Validation Scorecard

Subsystem	Result	Detail	Verdict
RoCE v2 bandwidth	98.18 Gb/s send/write, 98.17 read	Top of Broadcom Thor envelope; line rate	PASS
Full-duplex aggregate	194.99 Gb/s	190-196 typical; full line rate	PASS
RoCE latency (2 B msg)	write 2.41 / send 2.59 / read 4.39 us	Within published band	PASS
TCP sustained (back-to-back)	94.04 Gb/s, 0 retransmits	Top of the Linux kernel TCP envelope	PASS
Link & PCIe health	0 FCS/PCS/pause errors; AER clean	ibstat ACTIVE, Rate 100 both hosts	PASS
CPU efficiency (RoCE vs TCP)	9.2x less host CPU (RX)	RoCE ~1 core vs TCP ~6 cores at 100G	PASS
Through-switch (SONiC)	RoCE 98.18 Gb/s; TCP 94->68	TCP collapses without PFC (4.0M retransmits); RoCE robust	PASS
Power efficiency	RoCE saves 27-49 W/server vs TCP	~76 W per 100G link pair; ~1.5 kW/rack at 32 nodes	PASS



Hardware Inventory

System

Vendor	Tyrone (Netweb)
Model	srv218 (Camarero) <-> srv148 (MDA200A2N-224)
Hostname	srv218 / srv148

CPU

Model	srv218: 2x Xeon Gold 6338 (64C/128T) | srv148: 2x EPYC 9135 (32C/64T)
Sockets	2

Memory

Total	srv218: 128 GB DDR4-3200 | srv148: 512 GB DDR5-5600
Type	DDR4 / DDR5

BIOS / Platform

Version	srv218 AMI L1.14B (2025-09-24) | srv148 AMI ES312AMS.205T8 (2026-03-26)

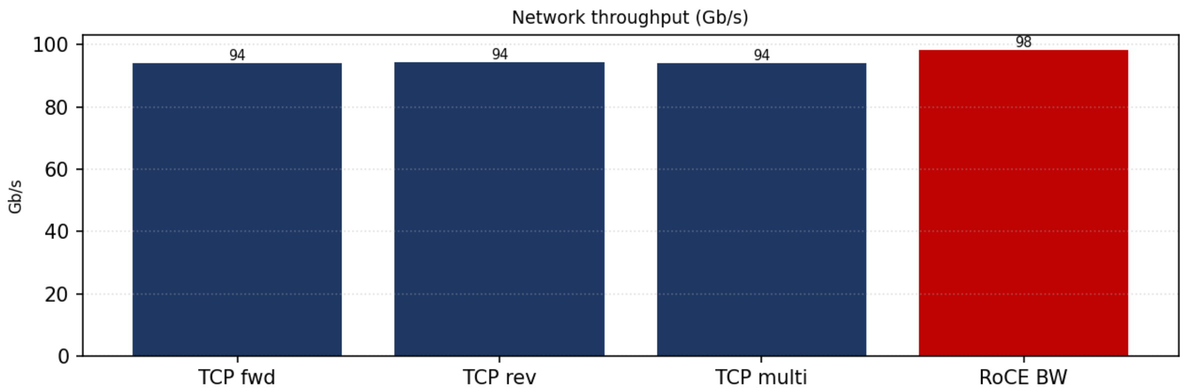
Kernel / OS

Distro	Ubuntu 22.04.5 LTS
Kernel	6.8.0-111-generic

Network Interfaces

Interface	Model	Speed	Driver
srv218 100G	Broadcom BCM57508 (200G card)	100 Gb/s (PCIe Gen4 x16)	bnxt_en/bnxt_re 226.0.145.1
srv148 100G	Broadcom BCM57504 (100G card)	100 Gb/s (PCIe Gen4 x16)	bnxt_en/bnxt_re 226.0.145.1
switch	Accton AS4630-54TE (SONiC 4.5.1)	100GBASE-CR, MTU 9100	no PFC (default lossy)

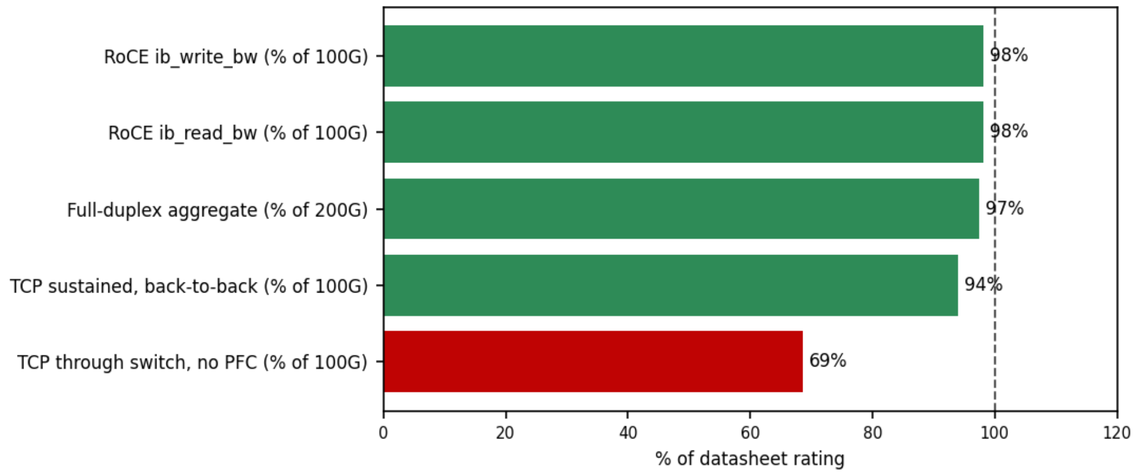
Network



TCP RX / TX (Gb/s)	94.04 / 94.34
RoCE ib_write_bw (Gb/s)	98.18
RoCE p99.9 latency (us)	8.56
MTU / PFC / ECN	9000 / off (back-to-back & test switch); enable for multi-hop TCP / off (test)
RDMA link	100 Gb/s, ACTIVE (ibstat Rate 100)
Peer / fabric	srv148 / back-to-back 100G DAC + Accton AS4630/SONiC switch

Specification Compliance

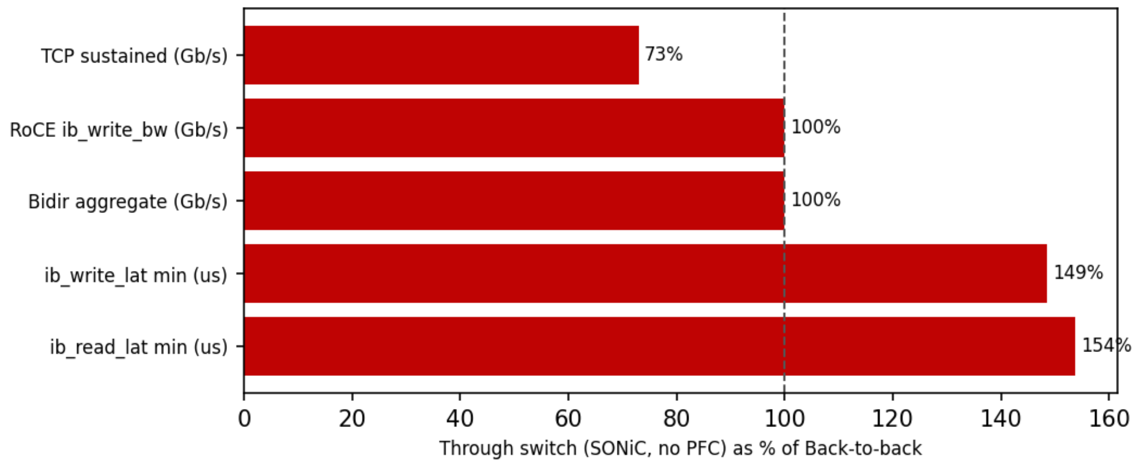
Measured performance as a percentage of datasheet rating. Reference line at 100%. Bars auto-colored: ≥90% green, 75-90% amber, <75% red.



Metric	Measured	Rated	Unit	% of rating
RoCE ib_write_bw (% of 100G)	98.18	100	Gb/s	98%
RoCE ib_read_bw (% of 100G)	98.17	100	Gb/s	98%
Full-duplex aggregate (% of 200G)	194.99	200	Gb/s	97%
TCP sustained, back-to-back (% of 100G)	94.04	100	Gb/s	94%
TCP through switch, no PFC (% of 100G)	68.62	100	Gb/s	69%

Cross-Server Comparison

Same NICs re-cabled through an Accton AS4630-54TE / SONiC switch (default lossy, no PFC).



Metric	Back-to-back	Through switch (SONiC, no PFC)	Delta
TCP sustained (Gb/s)	94.04	68.62	-27%

Metric	Back-to-back	Through switch (SONiC, no PFC)	Delta
RoCE ib_write_bw (Gb/s)	98.18	98.18	+0%
Bidir aggregate (Gb/s)	195.0	194.9	-0%
ib_write_lat min (us)	2.41	3.58	+49%
ib_read_lat min (us)	4.39	6.75	+54%

RoCE holds full line rate (98.18 Gb/s, 0 discards) through the switch; TCP collapses 94 -> 68 Gb/s without PFC (4.0M retransmits, 5.2M RX discards on CoS4). The switch adds ~1.2 us/hop for send/write and ~2.4 us round-trip for read (RDMA Read = request + response).

Component Health

Indicator	srv218	srv148	Verdict
FCS errors (TX+RX)	0	0	PASS
PCS symbol errors	0	0	PASS
RX / TX pause frames	0 / 0	0 / 0	PASS
RoCE port (ibstat)	ACTIVE, Rate 100	ACTIVE, Rate 100	PASS
PCIe AER	clean	clean	PASS

Kernel log (dmesg) diff: Back-to-back: 0 FCS / PCS / RX-align errors and 0 pause frames on both NICs; PCIe AER clean; RoCE ports ACTIVE at Rate 100. Through-switch TCP soak (no PFC): 4,021,988 retransmits and 5,215,543 RX discards (CoS4) with 0 pause frames - confirming switch buffer overrun without PFC, not a NIC fault (0 FCS/PCS/FEC errors, 0 link-downs). RoCE through the switch: 0 discards, 0 errors, full line rate.

Findings & Recommendations

Findings

- Both Broadcom Thor NICs operate at full 100 G line rate for RoCE v2 send/write/read (98.18 Gb/s), sustain 194.99 Gb/s full-duplex, with RDMA write latency 2.41 us - at the top of Broadcom's published envelope; ib_read_bw (98.17 Gb/s) exceeds the typical Thor 94-97 Gb/s window.
- RoCE uses ~9.2x less host CPU than TCP on the receive side (~1 core vs ~6 cores at 100 G); %softirq drops to 0% under RDMA - true kernel bypass. ~50+ CPU cores/node are recoverable across ten 100 G links by choosing RoCE.
- Through a production switch (Accton AS4630-54TE / SONiC, default lossy / no PFC) RoCE held full line rate (98.18 Gb/s, zero discards) while TCP collapsed 94 -> 68 Gb/s: 4.0M retransmits and 5.2M RX discards (CoS4) with no PFC pause frames to throttle the switch.
- Power: RoCE saves ~27-49 W per server vs TCP at sustained 100 G through the switch (~76 W per 100G link pair; ~1.5 kW continuous for a 32-node rack), measured via IPMI DCMI + turbostat. RoCE adds only 7-12 W above idle; TCP adds 39-55 W.
- The switch adds ~1.2 us each way (send/write) and ~2.4 us round-trip (read); CPU efficiency holds - RoCE still uses 7.7x less host CPU than TCP through the switch. Link health is perfect throughout (0 FCS/PCS/pause errors, AER clean, ports ACTIVE Rate 100).

Recommendations

- Enable lossless RoCE (PFC + ECN + DCQCN) on shared switches before deploying TCP at 100 G - RoCE is robust without it, but TCP collapses (94 -> 68 Gb/s) on a default-lossy switch.
- Persist the tuning: L2 MTU 9000 + 10.10.10.x IP via netplan/nmcli; set hostnames with hostnamectl; persist NIC queue/ring settings via if-up.d or udev.
- Prefer RoCE for storage / AI / latency-sensitive traffic: full line rate through any reasonable fabric, ~9x less CPU, and measurable power/TCO savings vs TCP on the same hardware.

- Populate more DIMM channels (srv218: 2 of 16; srv148: 8 of 24) - more memory bandwidth helps DMA-heavy 100 G workloads.
- For MPI / OSU benchmarks, rebuild Open MPI against UCX >= 1.15 to fix the RoCE QP setup blocker (the only test not completed in this campaign).

Conclusion

Area	Outcome	Verdict
RoCE v2 bandwidth & latency	98.18 Gb/s send/write/read; 2.41 us write latency; top of envelope	PASS
Full-duplex & TCP (back-to-back)	194.99 Gb/s aggregate; 94.04 Gb/s TCP with 0 retransmits	PASS
Through-switch (SONiC, no PFC)	RoCE full line rate; TCP needs PFC/ECN (collapsed to 68 Gb/s)	PASS
Power / TCO	RoCE saves ~76 W per 100G link pair vs TCP (~1.5 kW/rack at 32 nodes)	PASS
CPU efficiency	RoCE 9.2x (b2b) / 7.7x (switch) less host CPU than TCP	PASS

Reproducibility Appendix

Exact tooling, kernel command line, BIOS state and a SHA-256 of the raw data bundle, so this run can be audited and reproduced.

Perftest	ib_send/write/read_bw + _lat
Iperf3	TCP/UDP
Sockperf	PPS + ping-pong
Diagnostics	ethtool -m/-S, lspci -vv (AER), rdma link, ibstat; power via ipmitool dcmi + turbostat PkgWatt
Switch	Accton AS4630-54TE, SONiC 4.5.1-Enterprise (100GBASE-CR, MTU 9100, no PFC)
Driver Fw	bnxt_en / bnxt_re 226.0.145.1
Distro	Ubuntu 22.04.5 LTS
Kernel	6.8.0-111-generic
Tuning	MTU 9000, tcp rmem/wmem 256 MB, NIC rings 2047, combined queues 32, CPU governor performance, NUMA-pinned, irqbalance off
Campaign Window	back-to-back + through-switch, 09 Jun 2026
Note	Reconstructed from the validated source report (Final v2, incl. Section 11 through-switch + power) into the CoreBench results.json contract.